

# Paper I-A — Coined-Land Hinge-Type Adaptive Tray (HyTray-A): Integral Living-Hinge Flap with Criss-Cross Interlaced Stiffness Installation

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Invention Disclosure / Provisional Specification Draft — HyTray Series HT-01A

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## Abstract

A cross-flow distillation tray is disclosed in which vapour passages are closed by flaps that pivot about a **coined living hinge** — a narrow thinned land formed integrally in the deck plate by coining (plastic reduction of thickness). The flap is lanced on three sides and the fourth side coined to a defined thickness ratio  $t_h/t_d = 0.45\text{--}0.65$ , creating a pivot zone about which the flap rotates in direct response to vapour pressure. Flaps are provided in two to four stiffness families (crack pressures 1.0–7 mbar) whose spatial ratio across the deck sets a continuous open-area schedule  $A_o(U_s)$ , achieving 12–15:1 turndown with no loose parts. Separately, a **column assembly claim** is disclosed for the criss-cross interlaced installation: alternate trays have their stiffness families mirror-inverted, forcing vapour to spread laterally between stages and suppressing vapour channelling in large-diameter columns. The coined living hinge is lower-cost to tool than the leaf-spring and lip-seal variants, at the cost of higher concentrated fatigue stress at the pivot.

*Status note: all performance figures are model predictions pending CFD/rig validation. The coined-hinge approach concentrates bending stress in a short, work-hardened land; its fatigue endurance is inferior to the leaf-spring embodiment (HT-01B) but may be adequate at moderate chatter amplitudes verified by coupon testing.*

## 1 The Coined Living Hinge — What It Is and Is Not

The hinge land (10) is NOT a separate pivot, pin, or bearing. It is a zone of the deck plate itself that has been thinned by coining (a controlled plastic reduction hit, analogous to the living hinge of a polypropylene bottle cap). When vapour pressure acts on the underside of the flap, the flap rotates about this thin zone; when pressure drops, elastic recovery closes the flap. The hinge carries no torque to a separate component — the load path is entirely within the monolithic deck panel.

The operating angle is *not fixed*: it increases continuously with vapour rate. The *rest angle* (zero vapour) is set by the degree of pre-set during coining; all flaps of the same family in a given panel have the same rest angle.

## 2 Mechanical Character and Fatigue Limitation

The coined land concentrates rotation in a length  $\ell_h \ll L_{flap}$ . Surface bending stress:

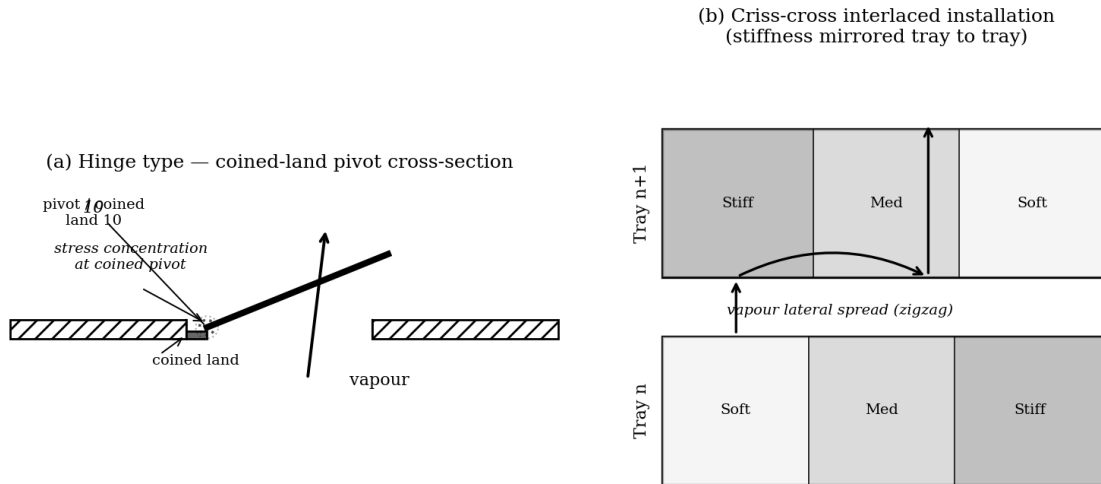
$$\sigma_a \approx \frac{E t_h \theta_{max}}{\ell_h},$$

which at  $t_h = 1.4$  mm,  $\ell_h = 3$  mm,  $E = 193$  GPa,  $\theta_{max} = 24^\circ$  gives  $\sigma_a \approx 580$  MPa — near or above austenitic SS yield. The notch factor  $K_f \approx 2$  at the coined-to-uncoined transition makes this worse. Coining removes the shear burr (the primary crack initiator) and introduces compressive residual stress that partly offsets the notch. Practical consequence: chatter amplitude must be controlled (deck-layout and vapour-path acoustic design) and a fatigue coupon programme is mandatory. The leaf-spring embodiment (HT-01B) distributes the same rotation over a much longer leg and is mechanically preferable where tooling complexity is acceptable. *No hole is placed in the hinge land*: a drilled hole is a stress raiser at the highest-stress location and is expressly prohibited.

The minimum open area is the perimeter shear gap  $g \approx 0.15\text{--}0.25$  mm (lance clearance), giving  $A_{o,min} = g(2L_f + w_f)$  per flap — the minimum orifice without any drilled hole.

## 3 Brief Description of the Drawings

Figure 1: (a) cross-section through one coined-land flap showing the pivot stress concentration and its contrast to the leaf-spring type; (b) criss-cross interlaced layout viewed from above — two adjacent trays with mirror-inverted stiffness families and the resulting vapour zigzag path.



**Figure 1:** (a) Coined-land pivot (10): stress is concentrated in a short thinned zone, returned by elastic recovery. (b) Criss-cross installation: soft/medium/stiff families mirrored between tray  $n$  and tray  $n + 1$ , producing lateral vapour spread between stages.

## 4 Criss-Cross Interlaced Installation

In the preferred multi-tray arrangement, the stiffness family layout of tray  $n$  is mirror-inverted on tray  $n + 1$  (and restored on tray  $n + 2$ , alternating). At partial load:

1. Vapour rises through the soft family on tray  $n + 1$  (low  $\Delta P_{cr}$ , left half in Fig. 1b).
2. It encounters the *stiff* family on tray  $n$  directly above (high  $\Delta P_{cr}$ , also left half) and is resisted.

3. Vapour deflects laterally and rises through the soft family of tray  $n$  on the *right* half.
4. The net path is a zigzag, increasing vapour lateral mixing between stages.

At full load all families are open and the layout has negligible effect on total  $\Delta P$ . The benefit accrues at partial load where vapour channelling is worst. This mechanism is independent of and additive to the GRADEX (HT-03) liquid-flow improvement.

**Parametric benefit (order of magnitude):** Vapour distribution coefficient  $\xi_V = \sigma_{Cs}/\bar{C}_s$  (fractional standard deviation of local capacity factor across the deck) is modelled to fall from  $\sim 0.15$  (uniform layout) to  $\sim 0.07$  at 50% load with alternating installation; efficiency recovery  $\approx E_{OG} \cdot [\xi_V^2 (\partial E_{MV}/\partial \xi_V)]$  to be calibrated by CFD.

## 5 Design Parameter Envelope

| Parameter                              | Claimed band        | Preferred          |
|----------------------------------------|---------------------|--------------------|
| Coined land thickness ratio $t_h/t_d$  | 0.45–0.65           | 0.55               |
| Coined land length $\ell_h$            | 2–5 mm              | 3 mm               |
| Flap plan size                         | 25–40 × 30–50 mm    | 32 × 40 mm         |
| Stiffness families (crack $\Delta P$ ) | 2–4 / 1.0–7 mbar    | 3 / 1.2, 3, 6 mbar |
| Spatial ratio (soft/med/stiff)         | 50/30/20 ±10        | 50/30/20           |
| Rest angle (set at coining)            | 5–15°               | 8°                 |
| Max operating lift angle               | 18–30°              | 24°                |
| Shear gap $g$                          | 0.15–0.25 mm        | 0.20 mm            |
| $A_{o,min}$ (shear gap)                | 3–6 %               | 4 %                |
| Criss-cross mirror period              | 1 (every tray) or 2 | 1                  |

## 6 Claims (draft)

**Claim 1.** A cross-flow contacting tray comprising a deck plate having a plurality of flap valves, each flap being formed integrally with the deck plate by lancing three sides of a rectangle and coining the fourth side to a thickness ratio  $t_h/t_d$  of 0.45–0.65 to form a living-hinge pivot land, the flaps being provided in at least two stiffness families differing in coined-land geometry such that the families crack open sequentially with increasing vapour load and the aggregate effective open area varies continuously between 3–6% and 14–18% of bubbling area.

**Claim 2.** The tray of claim 1, wherein the hinge land is free of holes, and the minimum open area is provided by the perimeter shear gap  $g = 0.15$ –0.25 mm of the lanced flap perimeter.

**Claim 3.** The tray of claim 1, wherein the coined land length is 2–5 mm and a compressive residual stress is induced by the coining operation to offset fatigue notch effects.

**Claim 4.** A column assembly comprising a plurality of trays according to claim 1, wherein the stiffness family spatial layout of each tray is the mirror inverse of the adjacent tray above or below, so that at partial vapour load, vapour rising through a low-resistance zone of one tray encounters a high-resistance zone of the tray above and is deflected laterally before ascending, producing cross-stage vapour spreading.

**Claim 5.** The assembly of claim 4, wherein the mirror inversion alternates on every tray, and at full vapour load the arrangement has no appreciable effect on total pressure drop.

**Claim 6.** A column of diameter greater than 2 m containing an assembly according to claims 4–5, the arrangement reducing the fractional standard deviation of vapour capacity factor across the tray by at least 30% at 50% of design vapour load.